

Mathieu Sze

mathsze@gmail.com | +1 (314) 327-1503 | Saint Louis, MO

EDUCATION

Saint Louis University

B.A. in Computer Science

B.A. in Mathematics, Statistics Concentration

Cumulative GPA: 3.61

Saint Louis, MO

Aug 2021 – May 2026

- Dean's List: Spring 2022, Fall 2023, Fall 2025
- Relevant Coursework: Software Engineering, Data Structures, Object-Oriented Software Design, Computer Organization Systems, Algorithms, Distributed Computing, Discrete Mathematics
- Member of the Math and Computer Science Club

SKILLS

Programming & Query Languages: SQL, Python, C, C++, Java

Data Analysis: Data Cleaning, Data Validation, Data Manipulation (Python, SQL), Statistical Analysis (coursework)

Tools & Technologies: GitHub, Git, Linux, Jupyter Notebook, Docker, CI/CD

Data Visualization: Power BI, Matplotlib,

Data Analysis Libraries: Pandas, NumPy

Business Tools: Microsoft Excel

Concepts: Client-Server Architecture, Networking, APIs, Data Documentation, Data Quality Validation

PROJECTS

Multithreaded Password Cracking Tool (C, Pthreads, Linux)

- Built a multithreaded password cracking tool in C using Pthreads and crypt_r to run brute-force hash matching in parallel
- Improved performance by spreading work evenly across threads and cutting out unnecessary repeated computations
- Tested different thread counts to see how performance scaled and where bottlenecks appeared

Network File Server & Client Application (C, Sockets, Linux)

- Built a client-server file system in C using sockets with asynchronous, non-blocking communication
- Created simple request/response flows to handle file retrieval and commands like ls and quit
- Made the system more reliable by adding error handling, input validation, and testing in a Linux environment

Collaborative Software Development (C/C++, Python, Git, CI/CD)

- Worked on a team project using Git for version control and CI/CD pipelines for automated testing
- Helped build features, fix bugs, and review teammates' code to improve overall quality
- Adapted quickly to new tools and workflows to meet project goals and deadlines

EXPERIENCE

Mathnasium

Math Tutor

Saint Louis, MO

Sep 2025 – Present

- Tutored elementary through high school students in mathematics, ranging from basic arithmetic to calculus.
- Improved students' problem-solving skills by breaking down complex concepts into manageable steps.

Saint Louis University

Student Researcher & Assistant

Saint Louis, MO

Aug 2022 – Present

- Completed specific research training and earned a CITI Certification.
- Collaborated with the program research assistant.
- Assisted in updating the International Faculty and Staff Association (IFSA) webpage.
- Teaching Assistant for Cura Personalis course

VOLUNTEERING

Culturally Appropriate Transformation for Culture of Health (CATCH) Program

Student Health Coach

Saint Louis, MO

Jan 2025 – May 2025

- Assisted participants in developing healthier lifestyle habits through personalized coaching.
- Conducted pre/post-assessments and health coaching sessions during the eight-week program.